Definite Integrals

Derivation 1 Finite Summation Formulas. First we need to discuss finite sums, as we will use them later in the section to calculate areas under curves. Let us start with the following summation formulas.

$$\sum_{k=1}^{N} 1 = \underbrace{1+1+\cdots+1}_{N \text{ times}} = N \tag{1}$$

$$\sum_{k=1}^{N} k = 1+2+3+\cdots+N = \frac{N(N+1)}{2} \tag{2}$$

$$\sum_{k=1}^{N} k^2 = 1^2+2^2+3^2+\cdots+N^2 = \frac{N(N+1)(2N+1)}{6} \tag{3}$$

$$\sum_{k=1}^{N} k^3 = 1^3+2^3+3^3+\cdots+N^3 = \frac{N^2(N+1)^2}{4} \tag{4}$$

Example 2 Calculate $1+2+3+\cdots+100$.

Solution.

$$1+2+3+\cdots+100 = \sum_{k=1}^{100} k = \frac{100(100+1)}{2} = \frac{10100}{2} = 5050.$$

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Example 3 Calculate $1^2+2^2+3^2+\cdots+10^2$.

Solution.

$$1^2 + 2^2 + 3^2 + \cdots + 10^2 = \sum_{k=1}^{10} k^2 = \frac{(10)(10+1)(2(10)+1)}{6} = \frac{(10)(11)(21)}{6} = 385.$$

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Derivation 4 Finite Summation Properties. How do we combine these together to find something like $\sum_{k=1}^{N}(k^2 + 2k)$?

If f and g are functions and c is a real number, then

$$\sum_{k=1}^{N} cf(k) = c\sum_{k=1}^{N} f(k) \tag{5}$$

$$\sum_{k=1}^{N}\big(f(k) \pm g(k)\big) = \sum_{k=1}^{N} f(k) \pm \sum_{k=1}^{N} g(k) \tag{6}$$

Example 5 Determine the value of $\sum_{k=1}^{N}(k^2 + 2k)$.

Solution.

$$\begin{aligned}
\sum_{k=1}^{N}(k^2 + 2k) &= \sum_{k=1}^{N} k^2 + \sum_{k=1}^{N} 2k \\
&\quad \text{by the sum-addition property} \\
&= \sum_{k=1}^{N} k^2 + 2\sum_{k=1}^{N} k \\
&\quad \text{by the scalar-multiple property} \\
&= \frac{N(N+1)(2N+1)}{6} + 2\left(\frac{N(N+1)}{2}\right)
\end{aligned}$$

by the first-power and square-sum formulas

$$= \frac{N(N+1)(2N+1)}{6} + \frac{6N(N+1)}{6}$$
$$= \frac{N(N+1)(2N+1+6)}{6}$$
$$= \frac{N(N+1)(2N+7)}{6}.$$

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Derivation 6 Approximating Area under a Curve. Why are we talking about this? How could we find the area under a curve?

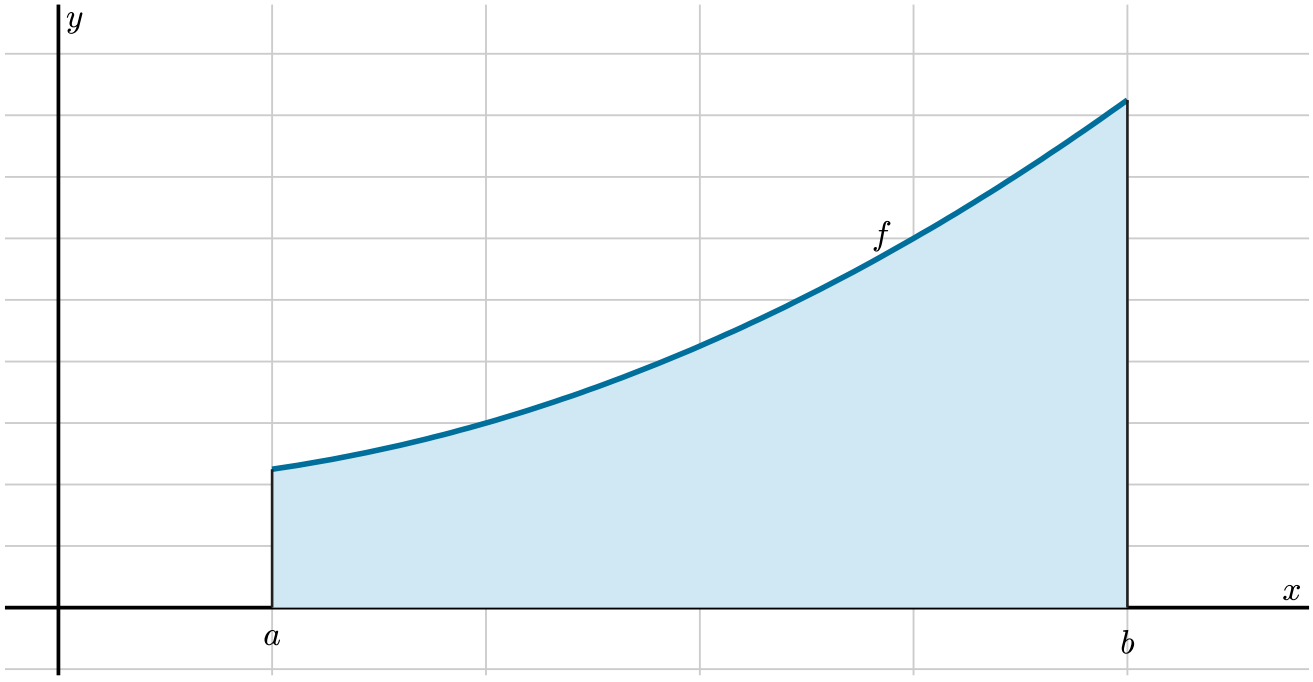

Figure 7 A generic region under a curve.

How might we approximate this area?

Use N rectangles of width $\Delta x = \frac{b-a}{N}$, with right-end sample points $x_k = a + k\Delta x$. Then

$$\text{area} \approx f(x_1)\Delta x + f(x_2)\Delta x + \cdots + f(x_N)\Delta x = \sum_{k=1}^{N} f(x_k)\Delta x,$$

which is a finite sum.

Derivation 8 The Definite Integral. How do you make the approximation better? More rectangles imply that we take a limit.

Definition 9 Definite Integral. The **definite integral** of f over $[a, b]$ is

$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\left[\sum_{k=1}^{n} f(x_k)\Delta x\right].$$

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Example 10 Use the definition of the definite integral to find the area under $4x^3$ on $[0, 3]$. In other words, calculate $\int_0^3 4x^3\,dx$.

Solution. First, calculate the width Δx and sample points x_k for n rectangles:

$$\Delta x = \frac{3-0}{n} = \frac{3}{n}, \qquad x_k = 0 + k\left(\frac{3}{n}\right) = \frac{3k}{n},$$

now, calculate the function values $f(x_k)$:

$$f(x_k) = 4\left(\frac{3k}{n}\right)^3 = \frac{108k^3}{n^3}.$$

Putting this into the definition:

$$\begin{aligned}
\int_0^3 4x^3\,dx &= \lim_{n\to\infty}\sum_{k=1}^{n}\left(\frac{108k^3}{n^3}\right)\left(\frac{3}{n}\right)\\
&= \lim_{n\to\infty}\sum_{k=1}^{n}\frac{324k^3}{n^4}\\
&= \lim_{n\to\infty}\left[\frac{324}{n^4}\sum_{k=1}^{n}k^3\right]\\
&= \lim_{n\to\infty}\left[\frac{324}{n^4}\cdot\frac{n^2(n+1)^2}{4}\right]\\
&= \lim_{n\to\infty}81\left(1+\frac{2}{n}+\frac{1}{n^2}\right)\\
&= 81.
\end{aligned}$$

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Example 11 Evaluate $\int_2^4 (2x-6)\,dx$ using the definition of the definite integral.

Solution. First, calculate the width Δx and sample points x_k for n rectangles:

$$\Delta x = \frac{4-2}{n} = \frac{2}{n}, \qquad x_k = 2 + k\left(\frac{2}{n}\right) = 2 + \frac{2k}{n},$$

now, calculate the function values $f(x_k)$:

$$f(x_k) = 2\left(2 + \frac{2k}{n}\right) - 6 = \frac{4k}{n} - 2.$$

Putting this into the definition:

$$\begin{aligned}
\int_2^4 2x - 6\,dx &= \lim_{n\to\infty} \sum_{k=1}^{n} \left(\frac{4k}{n} - 2\right)\left(\frac{2}{n}\right) \\
&= \lim_{n\to\infty} \sum_{k=1}^{n} \left(\frac{8k}{n^2} - \frac{4}{n}\right) \\
&= \lim_{n\to\infty} \left(\frac{8}{n^2}\sum_{k=1}^{n} k - \frac{4}{n}\sum_{k=1}^{n} 1\right) \\
&= \lim_{n\to\infty} \left(\frac{8}{n^2}\frac{n(n+1)}{2} - \frac{4}{n}n\right) \\
&= \lim_{n\to\infty} \left(4\frac{n+1}{n} - 4\right) \\
&= \lim_{n\to\infty} \left(4\left(1 + \frac{1}{n}\right) - 4\right) \\
&= 4(1+0) - 4 \\
&= 4 - 4 \\
&= 0.
\end{aligned}$$

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Homework Problems. Use finite summation properties and finite summation formulas to eval-

uate the following finite summations.

1. $\displaystyle\sum_{k=1}^{180} 1$

Answer. 180

2. $\displaystyle\sum_{k=1}^{200} k$

Answer. 20100

3. $\displaystyle\sum_{k=1}^{30} k^2$

Answer. 9455

4. $\displaystyle\sum_{k=1}^{10} k^3$

Answer. 3025

5. $\displaystyle\sum_{k=1}^{n} (2k-1)$

Answer. n^2

6. $\displaystyle\sum_{k=1}^{n} (k^2+4)$

Answer. $\frac{2n^3+3n^2+25n}{6}$

7. $\displaystyle\sum_{k=1}^{n} (k^2-k)$

Answer. $\frac{n^3-n}{3}$

8. $\displaystyle\sum_{k=1}^{n} (k-2)^2$

Answer. $\frac{2n^3-9n^2+13n}{6}$

Use the definition of the definite integral to evaluate the following definite integrals.

9. $\displaystyle\int_{-1}^{4} x\,dx$

Answer. $\frac{15}{2}$

10. $\displaystyle\int_{-3}^{5} (2x-1)\,dx$

Answer. 8

11. $\displaystyle\int_{-1}^{3} (-5x)\,dx$

Answer. -20

12. $\displaystyle\int_{-3}^{0} (3-2x)\,dx$

Answer. 18

13. $\displaystyle\int_{0}^{4} 3x^2\,dx$

Answer. 64

14. $\displaystyle\int_{0}^{3} (x^2-1)\,dx$

Answer. 6

15. $\displaystyle\int_{-1}^{3} (x^2-x)\,dx$

Answer. $\frac{16}{3}$

16. $\displaystyle\int_{0}^{3} x^3\,dx$

Answer. $\frac{81}{4}$